

Oscar Jakacki

oscarjakacki927@gmail.com | jakackioscar.github.io | github.com/jakackioscar | linkedin.com/in/oscar-jakacki | Evanston, IL

EDUCATION

Northwestern University — B.S. Electrical Engineering *Evanston, IL*
McCormick School of Engineering | Expected 2028

SKILLS

Languages: Verilog, C, Python

Digital Design: RTL design and simulation, digital logic, FPGA bring-up, Vivado

Hardware: PCB design (Altium, KiCad, Fusion 360), op-amp signal conditioning, soldering and board rework

Test & Measurement: RF equipment, oscilloscopes, logic analyzers, custom test tooling in Python

Signal Processing: FFT, digital filtering, fixed-point math

Embedded Systems: ESP32, ARM Cortex-M, Spartan-7 FPGA, Raspberry Pi

Other: Git, Linux

EXPERIENCE

Silicon/RF Intern, Stealth IoT Hardware Startup *Jun 2026 – Sept 2026*
San Diego, CA

- Built Python tooling for hardware bring-up testing, including a GPS post-processing pipeline with signal filtering that improved location accuracy to within 20 m
- Developed a software-based spectrum analyzer in Python for RF interference analysis, used in place of dedicated bench equipment during hardware bring-up
- Configured and tested FPGA-based hardware alongside Linux test infrastructure, supporting integration between RF front-end and digital signal processing chains
- Worked hands-on across GPS, RF, and FPGA subsystems as part of a small, cross-functional embedded/RF hardware team

Undergraduate Researcher, Embodied Systems Lab *Mar 2026 – Present*
Northwestern University

- Designed op-amp signal conditioning circuitry and built the analog front-end PCB, from schematic to fabrication, for a TENG-based biosignal acquisition system integrated into smart glasses
- Proposed and validated a novel sensor placement approach for extracting heart rate and blood pressure signals via PPG
- Developed embedded C firmware on an XIAO ESP32 for real-time biosignal acquisition and on-device signal processing
- Collaborated with graduate researchers on sensor design, analog signal conditioning, and low-power wearable hardware integration

Electrical Team Member, Northwestern Solar Car *Jan 2026 – Present*
Evanston, IL

- Redesigned the battery management system with a teammate, transitioning from an active to a passive cell balancing architecture built around LTC3300 and LTC6813 ICs
- Debugged LTC6820/LTC6813 daisy-chain communication issues across the battery module stack, improving reliability of cell voltage monitoring
- Improved cell voltage uniformity and overall pack efficiency through the balancing redesign

Undergraduate Researcher, Pleshko Lab, Temple University *Dec 2024 – Sep 2025*
Philadelphia, PA — O-PTIR spectroscopy study on bone composition

PROJECTS

Music Genre Classifier on an FPGA

Verilog, Spartan-7, Vivado

- Designed a hand-written Verilog signal processing pipeline (1024-point FFT, 26-band mel filterbank, quantized neural network) with no processor, OS, or vendor IP
- Simulated and verified the RTL design in Vivado; debugged a real-hardware clock-mapping bug that testbench simulation missed
- Achieved 86.3% classification accuracy on held-out samples and 67% on live audio input

Web Server on a Microcontroller

C, ARM Cortex-M0+, OpenOCD

- Implemented an HTTP server on a PY32F030 microcontroller with no network hardware, using a uIP stack over SLIP tunneled through bit-banged SWD

Attenda Health — 2nd Place Overall, WildHacks 2026

Next.js, React, Supabase

- Built a mobile hospital call system with automatic voice transcription/translation and a real-time bed-status map synced to the queue

PUBLICATIONS

Dave, R.N., Prodanich, J., Lai, Y., Jakacki, O., Lin, S., Thoene, J., Alshurafa, N., & Arora, N. (2026). "GlasTENG: Self-Powered Triboelectric Nanogenerator based Sensing of Pulse, Jaw, and Upper Facial Activity from Everyday Glasses." *ACM ISWC 2026 (Notes and Briefs)*. Second author.

Jakacki, O., Kust, S., & Pleshko, N. (2025). "Characterization of O-PTIR Orientation Effects on Bone." *Temple University, Department of Bioengineering*. Poster presented at the Sci-X Conference, October 2025. First author.